

Advanced (Habanero)

- Q1.** Solve the system:  $x^2 + y^2 = 4$ ,  $x^2 - y^2 = -1$   
(A)  $(-1, -\sqrt{3})$   
(B)  $(1, \sqrt{3})$   
(C)  $(0, 0)$   
(D) No solution  
(E)  $(-\sqrt{3}, 1)$
- Q2.** What is the solution to the system:  $y = x^2$ ,  $y = -x^2 + 2$ ?  
(A)  $(0, 0)$   
(B)  $(1, 1)$ ,  $(-1, 1)$   
(C)  $(0, 2)$   
(D) No solution  
(E)  $(\sqrt{2}, 2)$
- Q3.** Solve the system:  $x^2 + y^2 = 16$ ,  $y = x^2 - 4$   
(A)  $(0, -4)$   
(B)  $(2, 0)$   
(C)  $(-2, 0)$   
(D)  $(1, -3)$   
(E)  $(-1, -3)$
- Q4.** The graphs of  $y = x^2 - 1$  and  $y = -(x - 2)^2 + 1$  intersect at what points?  
(A)  $(0, -1)$ ,  $(2, 1)$   
(B)  $(1, 0)$   
(C)  $(0, 0)$   
(D) No intersection  
(E)  $(1, 0)$ ,  $(3, 0)$
- Q5.** What is the number of solutions to the system:  $x^2 + y^2 = 25$ ,  $x^2 - y^2 = -7$ ?  
(A) 0  
(B) 1  
(C) 2  
(D) 4  
(E) Infinitely many
- Q6.** The system of equations  $y = x^2 - 4$ ,  $y = -(x + 2)^2 + 1$  has which of the following solutions?  
(A)  $(-2, 1)$   
(B)  $(-1, -3)$   
(C)  $(-1, 1)$   
(D)  $(0, -4)$   
(E)  $(1, -3)$
- Q7.** Solve the system:  $y = x^2$ ,  $y = (x - 1)^2 - 1$   
(A)  $(0, 0)$ ,  $(1, 1)$   
(B)  $(0, 0)$ ,  $(2, 1)$   
(C)  $(1, 1)$ ,  $(2, 1)$   
(D) No solution  
(E)  $(0, 0)$ ,  $(1, 0)$
- Q8.** The system of equations  $x^2 - y^2 = 3$ ,  $y = x^2 - 4$  has which of the following solutions?  
(A)  $(2, 0)$   
(B)  $(-2, 0)$   
(C)  $(1, -3)$   
(D)  $(-1, -3)$   
(E)  $(\sqrt{3}, -1)$

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations  $y = x^2 - 2x + 1$ ,  $y = -(x + 1)^2 + 4$  and provide a graphical representation.
- Q10.** Find the points of intersection between  $y = x^2 - 3$ ,  $y = (x + 2)^2 - 1$ , and interpret the solutions graphically.

Chef's Tips

- Tip 1: Ensure all calculations are shown
- Tip 2: Graphical solutions should include a sketch of the graphs with intersection points clearly labeled
- Tip 3: Students should be able to justify their solutions mathematically